Independant Component Analysis (ICA) and its non linear extension through VAEs

Adam Rozzio, Romain Cargnello, Arthur Barjot

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Author Contributions: Arthur developed the theoretical framework for both ICA and iVAE, with a particular attention on the theoretical motivation of ICA, Adam contributed to the theory and application of ICA, with a implementation and an experimentation of the method, and Romain focused on the theory and application of iVAE, he spends some time on the latent reconstruction and the MCC metric.

1 Introduction to independent component analysis (ICA)

This section is largely inspired by [1], it presents ICA in detail, and there are also some improvements compared to [1]: we will deeper explanations on some arguments, and we reformulate some motivations.

Let $A \in \mathcal{M}_n(\mathbb{R})$, and $T > 0$. The aim of ICA is to reconstruct a signal $s = (s_1, ..., s_n)^T : [0,T] \to \mathbb{R}^n$, from the knowledge of $x := As$ (in this section we only talk about linear ICA). More precisely, we consider x as a random variable over $\mathbb{R}^n$, and $x(t)$ will be a sample of this random variable, we will find A^{-1} in order to obtain $s = A^{-1}x$, we write (A, s) a theoretic solution. We make a fundamental hypothesis over s, which is that the s_i are independent. This hypothesis is not restrictive in practice, as we will see during the experiment.

1.1 Reduction of the problem

A first observation is that if we take $s_i \sim \mathcal{N}(m_i, \sigma_i^2)$, and $\Omega \in O_n(\mathbb{R})$, then Ωs is also a Gaussian vector with independent components, and using the only hypothesis that the s_i are independent, it is impossible to distinguish the solutions (A, s) and $(A\Omega^{-1}, \Omega s)$. Hence we have to assume that s is not a Gaussian vector. More generally, for any s with independent components, if we take $\Omega \in \mathcal{M}_n(\mathbb{R})$ a permutation matrix, or a scaling matrix, the same problem appears, $(A\Omega^{-1}, \Omega s)$ is also a solution. First for the scaling, we will assume the s_i are scaled, and then we are not going to give any attention to the sign of the s_i, or to their order.

1.2 Intuition of non-gaussianity

In this section, the argument will not be theoretic, but we will explain the intuition of the ICA for s with i.i.d. components (the components have the same law only in this section). We denote $W := A^{-1} \in \mathcal{M}_n(\mathbb{R})$, and $W^T = \begin{pmatrix} w_1 & w_2 & \dots & w_n \end{pmatrix}$, such that $s_i = w_i^T x$. The idea behind ICA is that, thinking about the central limit theorem, the linear combination of i.i.d. random variables (r.v.), is closest to be a gaussian r.v. than the two initial, we will say that the sum is more gaussian than the initial r.v. (we explain later how to quantify the gaussianity of a r.v.). Hence if we denote $y := w^T x$, where $w \in \mathbb{R}^n$, then $y = (A^T w)^T s$ is more gaussian than the s_i. Hence minimising the gaussianity of y result in finding $(A^T w)^T$ such that $y = (A^T w)^T s$ is one of the s_i, and it is exactly finding w equal to one of the w_i.

So we have the idea to find the w_i: we define $\phi(w) := w^T x$, and we maximise it, up to a metric of non-gaussianity, over the set of scaled r.v.. ϕ will have $2n$ local maximum, which are the $\pm s_i$. We can find the s_i iteratively because they are independent, therefore decorrelated, which correspond to a notion of orthogonality of the s_i in $L^2(\mathbb{R})$. Hence if you find one s_i, you will now optimise ϕ over $\{s_i\}^\perp$. But in practice the algorithm parallelize these steps and optimize all the vector s directly.

1.3 The theoretical argument

In this section we will explain the theoretical arguments that reveal non-gaussianity.
We define the entropy of a r.v. with density f by $H(y) := -\int f(y)\log(f(y))dy$, and the mutual information of a random vector $y = \begin{pmatrix} y_1 & \dots & y_n \end{pmatrix}^T$:

$$I(y) := \left(\sum_i H(y_i)\right) - H(y) \tag{1}$$

Indeed $I(y)$ result in the Kullback-Leibler divergence of y with $y_1 \otimes \cdots \otimes y_n$, hence the mutual information is positive, and zero if and only if the y_i are independent. Hence it is the theoretic quantity we need to optimize to solve the ICA problem, let us understand the link with non-gaussianity.
We define the negentropy: $J(y) := H(\nu_y) - H(y)$, where $\nu_y \sim \mathcal{N}(0, \sqrt{\mathbb{V}(y)})$. An important result is that for a fixed variance, gaussian variables maximized the entropy, hence J is positive and minimizing it remains in maximising the non-gaussianity: it is a measure of non-gaussianity.
Now, remark that as we optimize over scaled random variables $(y_1, \dots, y_n)$, finding y maximising $\sum_i H(y_i)$ is equivalent to finding it minimising $\sum_i J(y_i)$ because $H(\mathcal{N}(0, \sqrt{Var(y_i)})) = H(\mathcal{N}(0,1))$ is constant.
Then if $y = Wx$, we have $I(y) := (\sum_i H(y_i)) - H(x) - \log(|\det(W|)$. But if W is such that the y_i are scaled and independent (exactly what we research), then $\det(\mathbb{E}[Wx(Wx)^T]) = \det(I_n) = 1 = \det(W)^2 \det(\mathbb{E}[xx^T])$, if $\det(\mathbb{E}[xx^T]) \neq 0$ (we ensure this in the algorithm), we obtain that $|\det(W)|$ only depends on x the observation, hence it is a constant of the problem. So finding W, such that the y_i are scaled and maximising $I(Wx)$ (which aims to find the y_i independent) is equivalent to find it such that the y_i are scaled and maximising $\sum_i H(y_i)$, and this is equivalent, as said before, to maximise $\sum_i J(y_i)$.
What we have just shown is that the theoretical way to solve the ICA problem is to maximise the negentropy of the components y_i (ensuring they are always scaled).

1.4 Measure of non-gaussianity

There are few measures of non-gaussianity, but the general idea is to consider a function G and define for all centred and scaled r.v. y the cost $C : y \mapsto [\mathbb{E}[G(y)] - \mathbb{E}[G(\nu)]]^2$, where $\nu \sim \mathcal{N}(0, I_n)$.

First remark if G is quadratic, $C(y)$ always vanished by the hypothesis on y. Hence we consider G non-quadratic. Let us focus on particular cases:

- if $G = y^4$, we are dealing with the kurtosis, hence it is very simple to calculate but we can show that it is not a robust measure of non-Gaussianity.
- if we take $G = \log \circ f$ where f is the density of y. It is not a true possible G because in this case it depends on y, but suppose we can, then we find the square of the negentropy (used in the previous section). Hence it has theoretical guaranties, but it is very hard to compute, if we want to, we should approximate the density of y, for every y we are interested in.

Considering those particular cases, we understand that we need to choose G in order to combine the good properties of kurtosis and negentropy, it has to be simple to compute, and it has to approximate negentropy in order to have theoretical guaranties.

Indeed choosing G growing slowly is more robust, the following functions have proved to be good choices:

$$G_1 : u \mapsto \frac{1}{a}\log\cosh(au) \quad G_2 : u \mapsto -\exp(-u^2) \tag{2}$$

where $a \in [1, 2]$ is a suitable constant.

1.5 Numerical Experiments

In this section we display the results from the application of both the 'symmetric' and 'deflation' ICA algorithm, presented on algorithm 2 and algorithm 1 in appendix, on increasingly complex unimodal source separation tasks : Synthetic data, Open source data from different field, our voices recorded with our phone.

Synthetic data. We generated 3 synthetic sources : a square signal, a saw tooth signal and a sinusoidal signal. These were the three independent sources to separate once mixed.
In order to save some space for the main text we display the results of the recovered sources with the 'deflation' ICA in the Appendix. The two approaches lead to a very satisfying result as one can observe on Figure 1

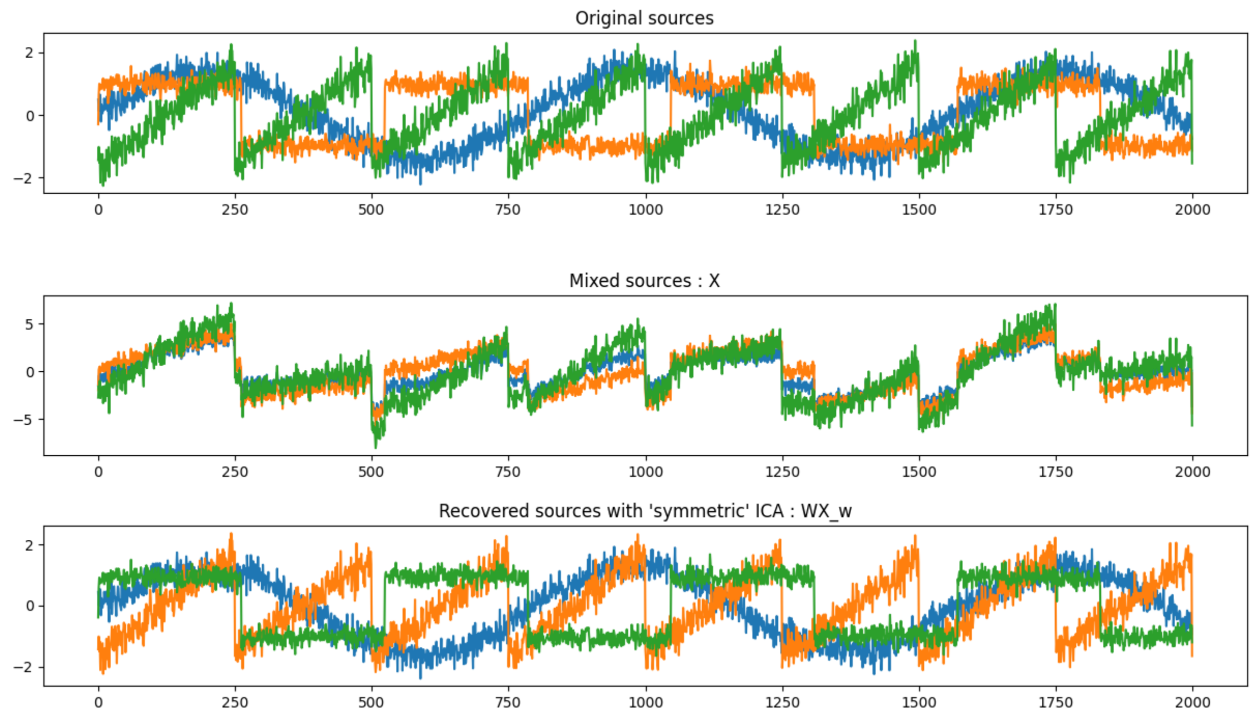


Figure 1: Original, mixed and recovered sources for synthetic data.

Open source data. Increasing the difficulty of the source separation task we consider time series coming from open source datasets imported from sktime.datasets with `load_airline`, `load_lynx` and `load_uschange`. The results are very good, you can find Figure 5 on appendix.

Our data. Finally, we assess the robustness of the method on real data. More precisely we recorded eight different audio which we mixed and managed to recover with a good quality the original independent sources. The Figure 6 in the appendix shows the original tracks, the mixed version and the algorithm gives a very good reconstruction of each track.

2 Non-linear ICA in the VAE framework

2.1 Introduction

Suppose now that $x = f(z)$ where z has independent components (in the VAE framework, the source s is denoted z) , but f is no longer a linear function. We just suppose that f is injective, so that $z = f^{-1}(x)$ is well define. In this framework, we no longer have to assume that x and z have the same number of components. To solve the problem with VAE, we consider z as a latent variable, the problem is that x and z are totally correlated, $p(x|z)$ is degenerated. Hence we introduce a noise ϵ with density p_ϵ, and the problem we will be interested in is to find z and f knowing x and p_ϵ where:

$$x = f(z) + \epsilon \tag{3}$$

Indeed we have: $p(x|z) = p_\epsilon(x - f(z))$, we denote it $p_f(x|z)$, f will be learn as a neural network. The problem now, which is general in non-linear ICA, is the lack of identifiability: in the linear case, supposing

z non-gaussian, with independent components and scaled was sufficient to reconstruct it, which is no longer true in non-linear ICA. We need more restrictive hypothesis on z, that is why, like in [2], we introduce an observed variable u and we suppose that z depends on u, with the following form for the density of z knowing u:

$$p_{T,\lambda}(z|u) = \prod_i \frac{Q_i(z_i)}{Z_i(u)} \exp\left(\sum_{j=1}^{k} T_{i,j}(z_i)\lambda_{i,j}(u)\right) = \prod_i p_{T_i,\lambda_i}(z_i|u)$$

Here the Q_i and the $T_{i,j}$ will be fixed, the $Z_i(u)$ are normalisation parameters, and the $\lambda_{i,j}(u)$ will be learned: we consider a neural network $u \mapsto \lambda(u)$.
Hence we have $p(x, z|u) = p_f(x|z) \times \prod_i p_{T_i,\lambda_i}(z_i|u)$, it can be represented by the graphical model Figure 2.

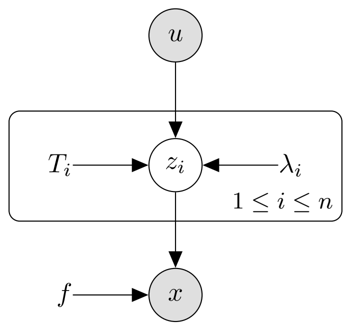


Figure 2: Probabilistic graphical model we used for iVAE.

2.2 Identifiability theory

The article [2] present in its forth parts some theorems about the identifiability of the model presented above. We will illustrate those result in a particular case. Suppose $\epsilon \sim \mathcal{N}(0, \sigma^2 I_n)$, $k = 2$, and $T_{i,1} : z \mapsto z$, $T_{i,2} : z \mapsto z^2$, for all i (it is the case study for the experiments). Then the hypotheses of theorems 1 and 3 of [2]: (i) the $T_{i,j}$ are twice differentiable everywhere, and (ii) for all i, $T_{i,1}$ and $T_{i,2}$ are linearly independent on every subspace of positive measure, are automatically verified. The other hypotheses are: (iii) there must be some $u_0, ..., u_{2n}$ such that $\begin{pmatrix}\lambda(u_1) - \lambda(u_0) & \cdots & \lambda(u_{2n}) - \lambda(u_0)\end{pmatrix} \in \mathbb{R}^{2n \times 2n}$ is invertible, and (iv) f should have all second order cross derivatives. (iv) is easy to achieve, as f is represented by a neural network, we just need to choose a smooth activation function. However, (iii) is hard to verify, but it is generally true. Indeed it is not verify only if $\lambda(\mathbb{R}^{2n})$ is contain in a strict affine subspace of $\mathbb{R}^{2n}$.

Under those hypotheses, the theorem 3 apply and we conclude as in the linear ICA that we can find the z_i up to a sign change of a permutation.

3 Experimentations

3.1 Dataset

3.1.1 Description of the dataset

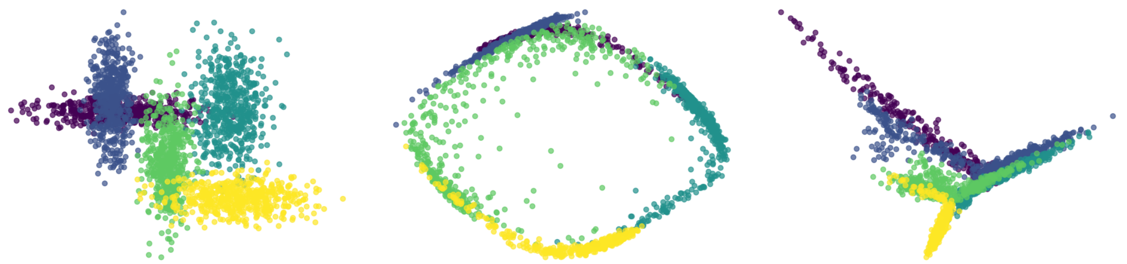
Following [2], the dataset is composed of: (i) independent sources s generated as non-stationary time series, (ii) a conditioning variable u defined as the one-hot segment index, (iii) signals x produced by a non-linear transformation of the sources.

The non-stationarity is crucial: u drives variations in the source distribution that enable identifiability. Each source is segmented into M blocks, each of length L, with segment-specific parameters. Since u identifies the segment, it provides exactly the statistical diversity required by the iVAE theorem.

In [2], they used sources s as non-stationary Gaussian time-series, and a Multi-Layer Perceptron (MLP) for obtaining x as a a non-lineartransformation of s.

3.1.2 Which dataset was actually used in the former article?

From [2], we know that the MLP to compute x from s is a 4-layer MLP, but we do not know which non-linearities, known as activation functions, were used in the MLP. In the article's GitHub repository, we can state that they used either Leaky-ReLU functions ($f(x) = \max(0, x) + s \min(0, x)$ with s the value of the slope for $x < 0$) or 'xtanh' functions ($f(x) = sx + \tanh(x)$ with s the slope value). Figure 3 shows the plot of the sources space and the signals space with both non-linearities (with $M = 5$, $L = 512$ and $d = 2$ sources). We can clearly see that the signals space using 'xtanh' non-linearities does look like the signals space present in the former article [2], and we can thus state that these non-linearities were used in the article.



(a) $p_{\theta^*}(z|u)$ (b) $p_{\theta^*}(x|u)$ with 'xtanh' activations (c) $p_{\theta^*}(x|u)$ with Leaky-ReLU activations

Figure 3: Visualization of $p_{\theta^*}(z|u)$ and $p_{\theta^*}(x|u)$ using different activation functions.

3.2 Implementation of the iVAE

In order to implement an iVAE, we need: (i) a decoder $p_f(x \mid z) = \mathcal{N}(x - f(z), \sigma^2 I)$ with $\sigma^2 = 0.01$, (ii) an encoder $q_\phi(z \mid x, u) = \mathcal{N}(g(x, u), \mathrm{diag}(\sigma^2(x, u)))$, (iii) a prior network that models $z \mid u$ via natural parameters produced by an MLP.

In the former article, the functional parameters of the decoder (f) and the inference model (g, σ2) as well as the conditional prior (λ) are chosen to be MLPs composed of 3 to 6 layers, hidden sizes between 10 and 200, and Leaky-ReLU activations (with a slope of 0.1). This last point is surprising because according to identifiability theory, f should be twice differentiable and thus be a MLP with twice differentiable activation functions, such as 'xtanh' activation functions.
In our experiments, we will use MLPs with 3 layers of size 128 each and (if not precised) Leaky-ReLU activations with a slope of 0.1.

3.3 The Mean Correlation Coefficient (MCC) metric

3.3.1 Definition

To quantify how well the inferred latent components z recover the true sources s, we use the *Mean Correlation Coefficient* (MCC) [3]. This metric measures the average absolute correlation between corresponding components, after optimally matching each inferred component to the true one:

$$\mathrm{MCC}(s, z) = \max_{\pi \in \mathcal{P}} \frac{1}{d} \sum_{i=1}^{d} \left| \mathrm{corr}\big(s_i,\ z_{\pi(i)}\big) \right| .$$

Here, $\mathcal{P}$ denotes all permutations of $\{1, \dots, d\}$, and the maximization accounts for arbitrary permutations and sign flips in the latent representation. MCC is a standard performance measure in non-linear ICA, as it assesses component-wise alignment independently of ordering or scaling ambiguities. A high MCC between the inferred latent components z and the true sources s means that we successfully identified the true parameters and recovered the true sources, up to point-wise transformations.

3.3.2 A secret of high MCC values

Now that we have a dataset, an implemented model and a metric, we can train an iVAE and compare the latent representation with the true source space. Figure 4 shows the results of two training runs. We observe that the iVAE's latent space does not accurately reconstruct the true source space, even though the MCC values are relatively high (MCC $= 0.913$ and MCC $= 0.960$), so high MCC values (as obtained in [2]) can hide lack of reconstruction. It thus would have been interesting to see latent reconstructions and the associated MCC in the article, rather than only the latent representation or only the MCC. We can also notice that the second run has a higher MCC because its latent representation has begun to unfold toward the true source structure. Interestingly, the iVAE's latent space also globally resembles the signal space (Figure 3b), suggesting that the signal space itself may have a high MCC (≥ 0.8) with the true source space.

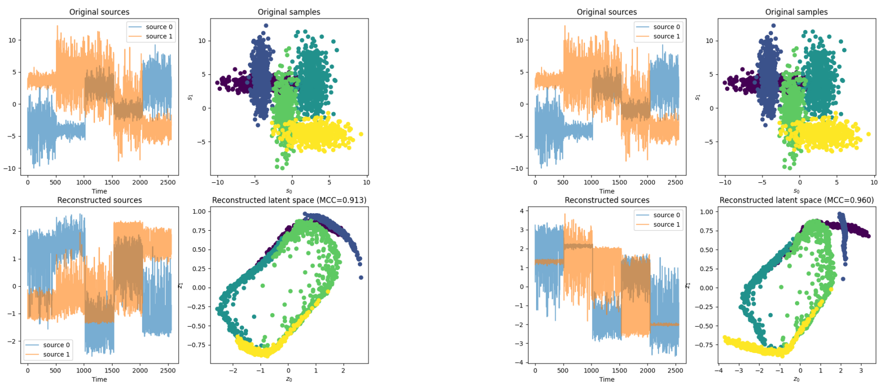


Figure 4: Results from two independent training runs of iVAE on the same dataset during 100 epochs.

3.3.3 Is the signal correlated to the source?

As suggests the previous sentence, the signal space may have a high MCC with the true source space, which is non-trivial since MCC captures linear correlations while x is a non-lineartransformation of s. However, even if a linear correlation exists between these spaces, we expect that it vanishes as the number of non-linear layers increases. Figure 7 in appendix shows how the MCC between the source space and the signal space changes as the number of non-linearlayers between x and s increases (we ran 10 simulations of s and x for each number of non-linear layers to have a box plot). For 'xtanh' non-linearities, the MCC remains relatively stable, which an unexpected behavior. In contrast, for Leaky-ReLU non-linearities, the MCC is high for shallow networks but drops sharply to near zero as the number of layers increases. These results indicate that, depending on the type of non-linearity, increasing network depth can have markedly different effects on the correlation between x and s.

3.3.4 Is iVAE more correlated to the source than the signal?

We have shown that, for the dataset considered, the correlation between x and s remains consistently high when using 'xtanh' activation functions, and is only significant for shallow networks when using Leaky-ReLU activations. This leads to the question of when the correlation between s and z becomes significantly stronger than that between s and x, i.e when an iVAE outperforms the identity function in terms of MCC. Figure 8 in appendix shows that, for our dataset, the iVAE latent space achieves significantly higher MCC values than the signal space only when the network has at least 10 non-linearlayers and uses Leaky-ReLU activations. Even in this regime, the correlation between z and s is moderate (around 0.4). In all other cases, the difference between MCC(z, s) and MCC(x, s) is negligible.

References

[1] E. Oja A. Hyvärinen. Independent component analysis: algorithms and applications. 2000.

[2] Ricardo Pio Monti Aapo Hyvärinen Ilyes Khemakhem, Diederik P. Kingma. Variational autoencoders and nonlinear ica: A unifying framework. 2020.

[3] Lindasalwa Muda, Mumtaj Begam, and Irraivan Elamvazuthi. Voice recognition algorithms using mel frequency cepstral coefficient (mfcc) and dynamic time warping (dtw) techniques. *arXiv preprint arXiv:1003.4083*, 2010.

Appendix

With the notations of the studied paper [1], we can write the symmetric ICA and the deflation ICA as follows.

Algorithm 1: Symmetric ICA

Input : Whitened data matrix X, Initial weights W_{init}, Iterations number N

Output: Separating matrix W

$g = \tanh(\cdot)$

$W = W_{init}/\|W_{init}\|_2$; `// Initial normalization`

for $i = 1, \ldots, N$ **do**

- $y = WX$;
- $(\beta_i)_i = -(\mathbb{E}\{y_i g(y_i)\})_i$;
- $(\alpha_i)_i = -(1/(\beta_i + \mathbb{E}\{g'(y_i)\})_i$;
- $W = W + \mathrm{diag}(\alpha)(\mathrm{diag}(\beta) + \mathbb{E}\{g(y)y^T\}W)$; `// Newton step`
- $W = (WW^\top)^{-1/2}W$; `// Symmetric Orthogonalisation`

return W

Algorithm 2: Deflation ICA

Input : Whitened data matrix X, Initial weights W_{init}, number of sources N_s, Tolerence ϵ

Output: Separating matrix W

$W = W_{init}$

$\lim = \epsilon + 1$

for $c = 1, \ldots, N_s$ **do**

- $w = W[c, :]^\top$; `// Extract current weight vector`
- $w = w/\|w\|_2$
- **while** $\lim > \epsilon$ **do**
  - $w_{old} = w$
  - $w_s = w^T X$
  - $w_{new} = \mathbb{E}\{X \cdot g(w_s)^\top\} - \mathbb{E}\{g'(w_s)\} \cdot w$; `// Newton step`
  - $w_{proj} = \sum_{j=1}^{c-1}(w_{new}^\top w_j)w_j$
  - $w_{new} = w_{new} - w_{proj}$; `// Orthogonal projection onto the subspace of found components`
  - $w_{new} = w_{new}/\|w_{new}\|_2$
  - $\lim = |w_{new}^\top w_{old}| - 1$
- $W[c, :] = w_{new}^T$

return W

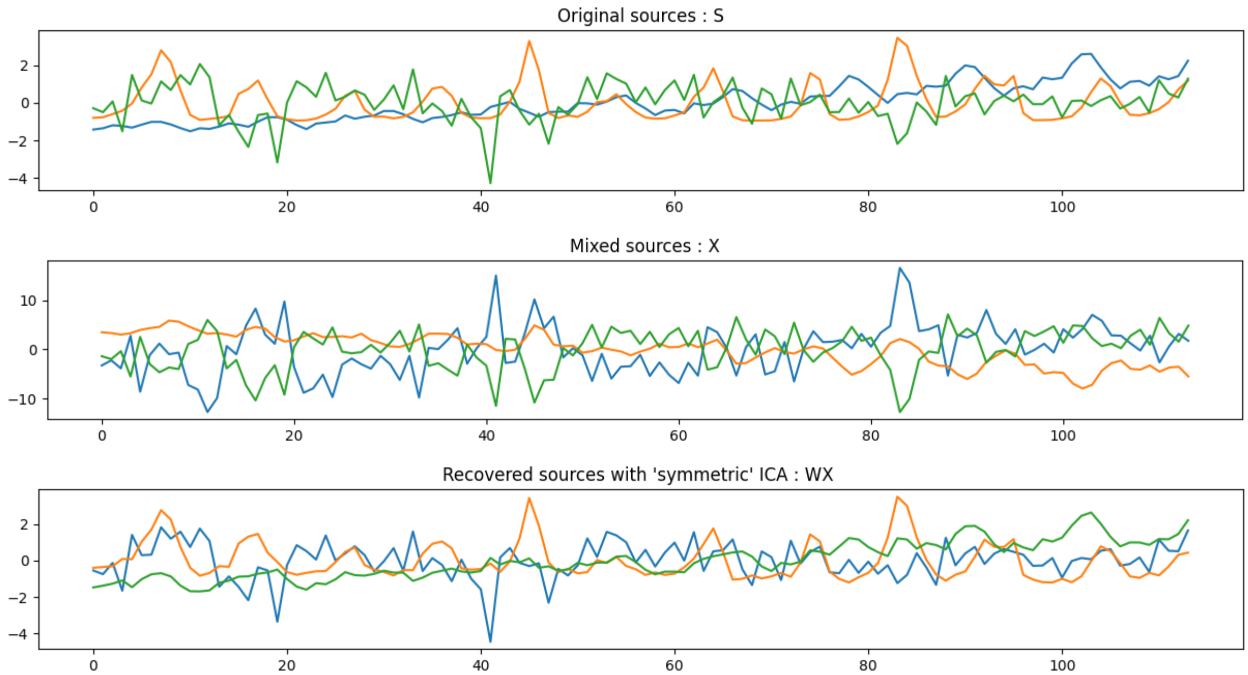


Figure 5: Original, mixed and recovered sources using ICA on open source data.

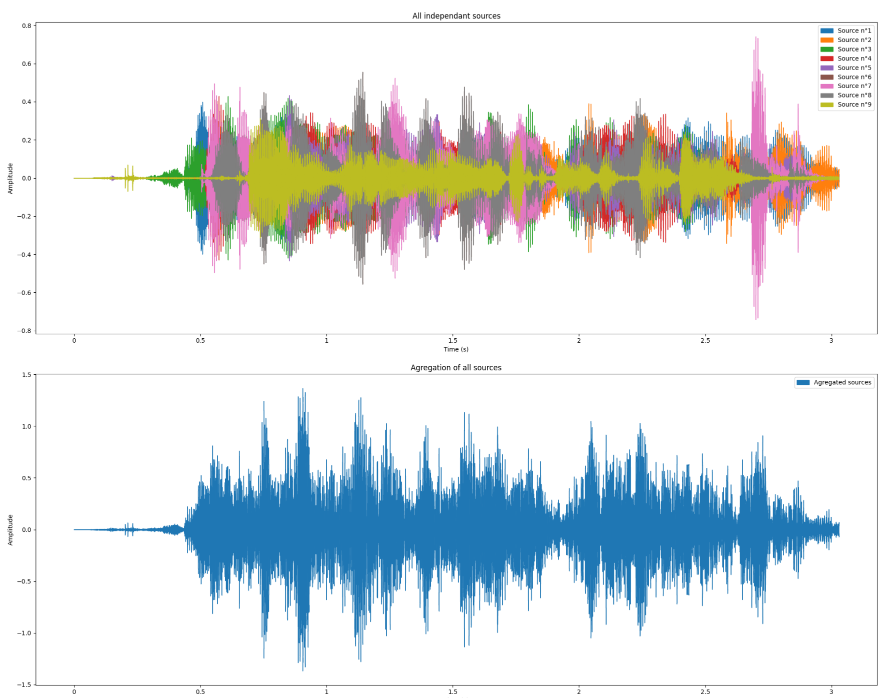
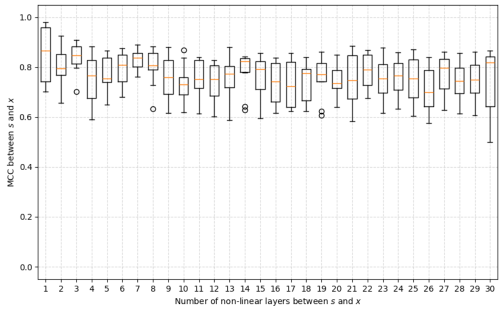
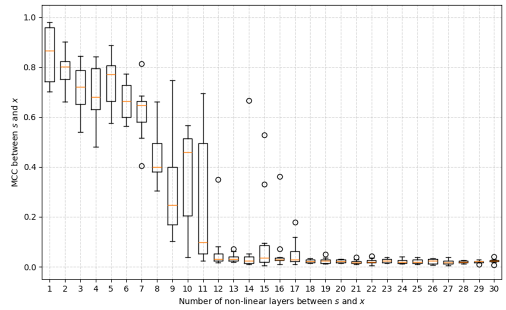


Figure 6: Original and mixed sources using ICA on our data.

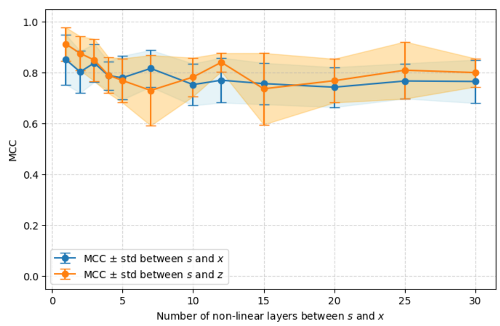


(a) $x = \text{MLP}(s)$ with 'xtanh' non-linearities

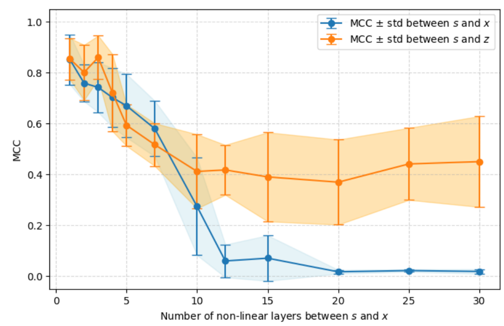


(b) $x = \text{MLP}(s)$ with Leaky-ReLU non-linearities

Figure 7: Effect of non-linearlayer depth on MCC between source and signal spaces.



(a) $x = \text{MLP}(s)$ with 'xtanh' nondata-linearities



(b) $x = \text{MLP}(s)$ with Leaky-ReLU non-linearities

Figure 8: MCC between sources and signals (s and x) and latents (s and z) as a function of non-linearlayer depth.